

DOMESTIC AIRLINES — TRAINING

Wind Limitations

Course handbook

Wind reports and components, AFM demonstrated crosswind and tailwind limits, crosswind by runway condition, and how to apply company limits.

3 lessons · 1 modules · about 0.5 hours

Training material only. It does not replace the Aircraft Flight Manual, FCOM, QRH, MEL or the Domestic Airlines Operations Manual, which always take precedence.

Contents

MODULE 1 — WIND AND LIMITATIONS

01 Wind Reports and Wind Components

02 Aircraft Wind Limitations

03 Company Limits and Crew Application

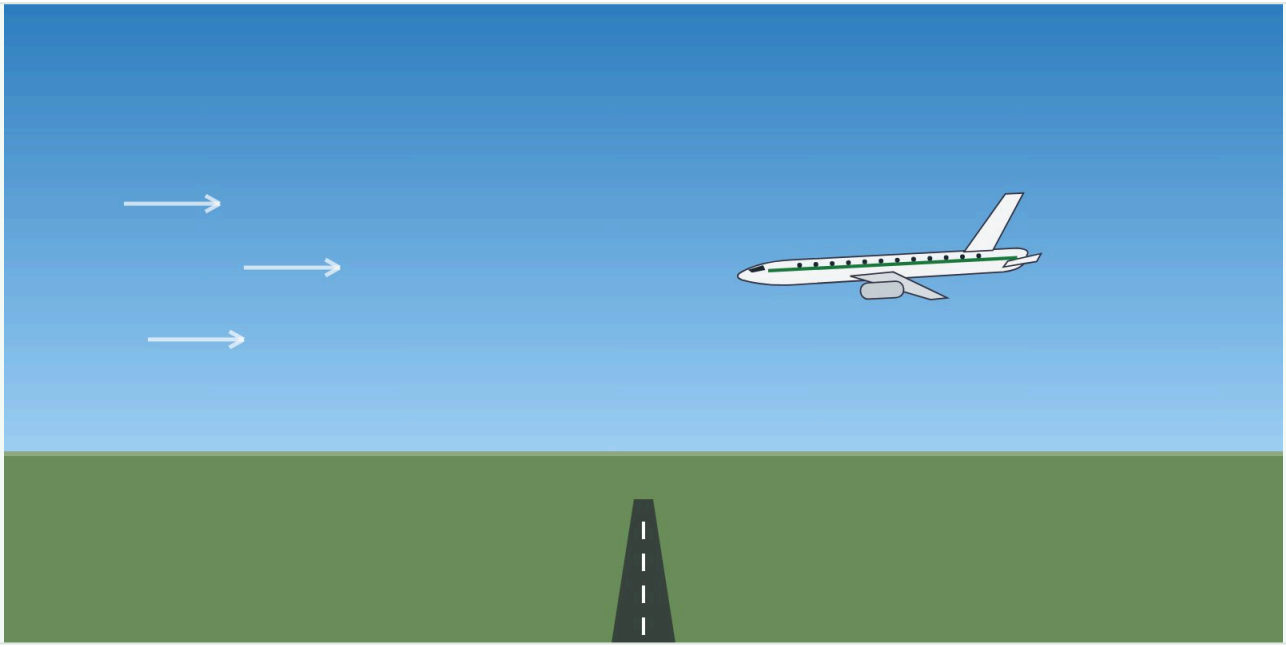
MODULE 1 · WIND AND LIMITATIONS

Lesson 01: Wind Reports and Wind Components

Lesson 1 of 3 · about 6 min

LEARNING OBJECTIVES

- Interpret wind reports correctly.
- Calculate headwind, tailwind and crosswind components.
- Use gusts correctly.

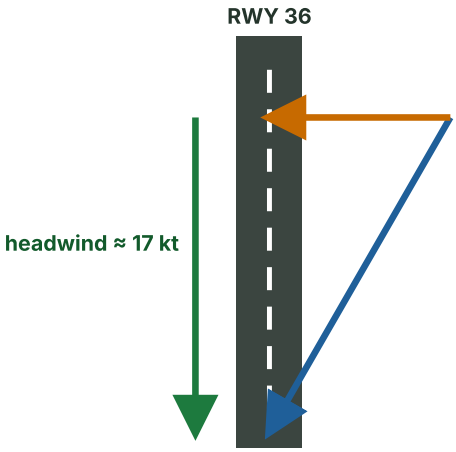


Calculate the components with the gust.

Wind reports

- In the **METAR**, the wind is a 10-minute mean; in reports for **take-off and landing** from the tower (and ATIS), it is a 2-minute mean for the runway (ICAO Annex 3).
- **Gusts** are reported when the maximum wind exceeds the mean by 10 kt or more: `27015G28KT` = 270°, 15 kt, gusting 28 kt.
- Variable direction: `27015KT 240V300` — direction varies between 240° and 300°.
- Directions in METAR / TAF are **true**; tower and ATIS winds are **magnetic**.

Components



Rule of thumb (crosswind)

30° off the runway → ½ of the wind
45° → about 0.7 of the wind
60° → about 0.9 of the wind
90° → all of the wind

crosswind = wind × sin(angle)
headwind = wind × cos(angle)

Always use the gust value when checking a crosswind limit, unless the company procedure states otherwise.

- Use the FMS / EFB, the wind component chart or the rule of thumb.
- For limits, check the crosswind with the **gust** and the tailwind with the steady wind and the gust as your procedure states.
- With a variable wind direction, use the most unfavourable direction.

KEY TAKEAWAYS

- METAR wind: 10-min mean, true; tower / ATIS: 2-min mean, magnetic.
- Gusts reported when 10 kt or more above the mean.
- Crosswind ≈ wind × sin(angle); use the gust for crosswind limits.

KNOWLEDGE CHECK

Wind 090°/20 kt on runway 06. Approximately what is the crosswind?

A. 5 kt
B. 10 kt
C. 17 kt
D. 20 kt

Answer: B. 30° off the runway: half the wind, about 10 kt.

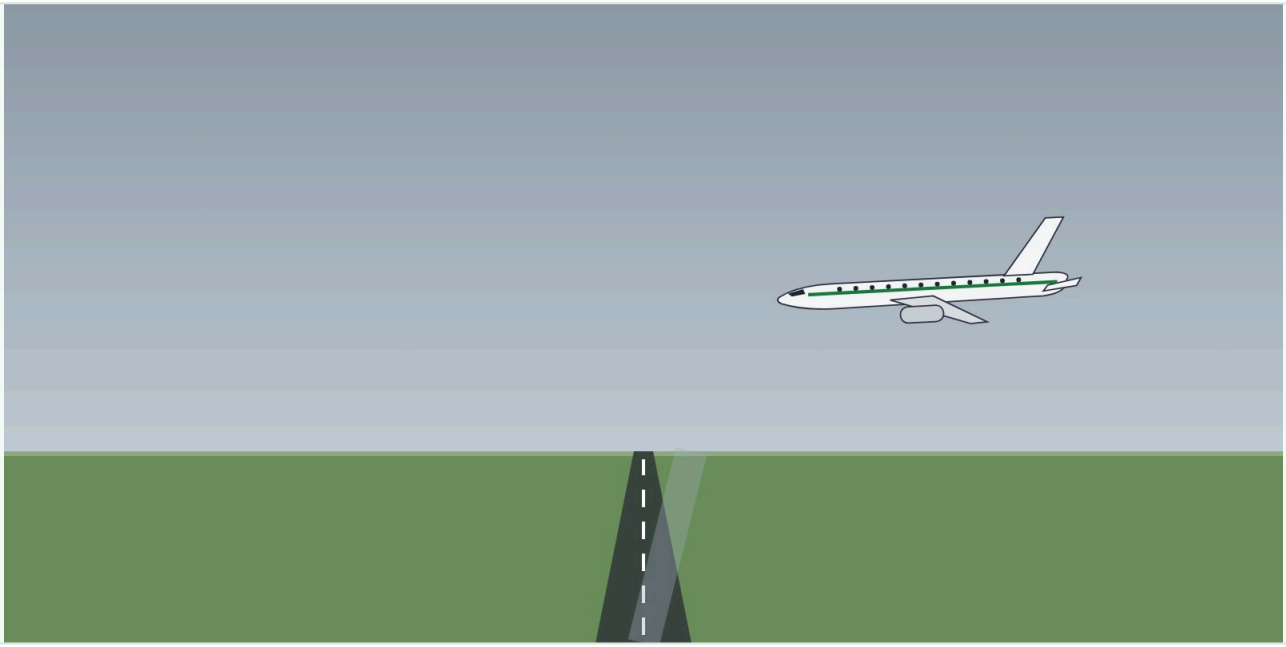
MODULE 1 · WIND AND LIMITATIONS

Lesson 02: Aircraft Wind Limitations

Lesson 2 of 3 · about 6 min

LEARNING OBJECTIVES

- Distinguish demonstrated crosswind from a limitation.
- Apply tailwind limits.
- Use the manufacturer's crosswind guidance by runway condition.



Crosswind limits decrease as the runway condition worsens.

Demonstrated crosswind

The AFM states the **maximum demonstrated crosswind** — the highest crosswind tested during certification. Unless the AFM says it is a limitation, it is **not** a structural or legal limit — but the company normally turns it, or a lower value, into an operational limit in the OM B.

Tailwind

The AFM gives a **maximum tailwind** for take-off and landing (commonly 10 kt; some aircraft are approved for more with specific conditions). This **is** a limitation. Performance must also be calculated with the reported tailwind.

Crosswind and runway condition

- Manufacturers publish recommended maximum crosswinds that **decrease as the runway condition worsens** — by runway condition code (RWYCC) or braking action.
- On a contaminated runway, use the crosswind for the reported RWYCC (see the Cold Weather Operations course).

Other wind-related limits

- **Autoland and low-visibility operations** have their own headwind, tailwind and crosswind limits.
- Doors: maximum wind for opening or closing passenger and cargo doors.
- Engine start in strong tailwind or crosswind (especially for some engines and propellers).

COMPANY LIMITS

The actual Domestic Airlines wind limits for each fleet — crosswind by runway condition, tailwind, gusts, limits for the first officer as pilot flying and for low-visibility / autoland — are in the **OM B**, the **FCOM / AOM** and the **AFM**. Always use them; the figures in this course are examples only.

KEY TAKEAWAYS

- Demonstrated crosswind: tested, not normally a limitation — but the company may make it one.
- Tailwind limit in the AFM is a limitation.
- Crosswind limits reduce with the runway condition; autoland has its own limits.

KNOWLEDGE CHECK

The AFM maximum demonstrated crosswind is:

A. Always a structural limit

B. The highest crosswind tested in certification — not a limitation unless the AFM says so

C. The same as the tailwind limit

D. Only for take-off

Answer: B. It is a demonstrated value, not a limitation unless stated.

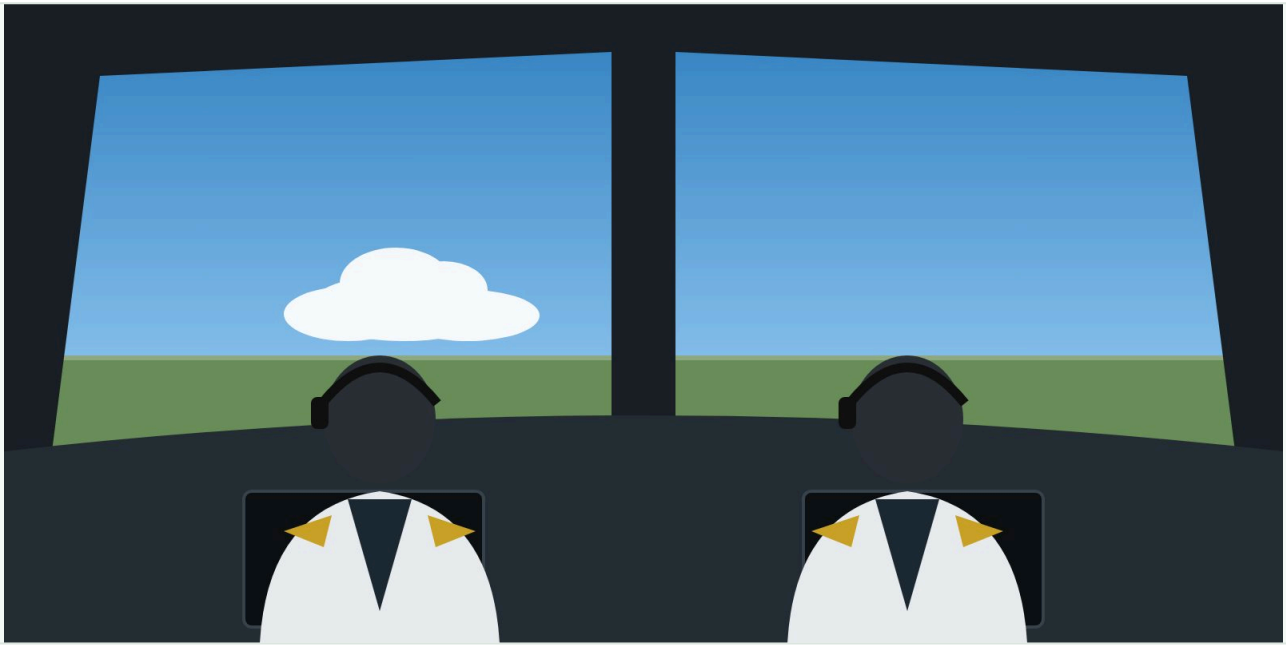
MODULE 1 · WIND AND LIMITATIONS

Lesson 03: Company Limits and Crew Application

Lesson 3 of 3 · about 6 min

LEARNING OBJECTIVES

- Apply company wind limits for each fleet and crew situation.
- Use speed additives and technique for wind.
- Decide when the wind does not allow take-off or landing.



Use the most restrictive limit that applies.

Company limits

The OM B normally defines, for each fleet:

- maximum crosswind for take-off and landing — dry, wet and by RWYCC — including gusts;
- maximum tailwind;
- more restrictive limits when the **first officer is pilot flying** or for crews with limited experience on type;
- limits for low-visibility operations and autoland;
- possible restrictions at specific aerodromes (OM C).

Application

1. Get the latest wind (ATIS / tower) and the runway condition.
2. Calculate the components with the gust; check the most restrictive limit that applies (aircraft, company, crew, runway condition, LVO).
3. Apply the approach speed wind additive defined in your FCOM / AOM.

4. Brief the crosswind technique and the go-around if the limits are exceeded on short final.

DECISION

If the wind exceeds a limit, do not take off or land — wait, use another runway or divert. Wind limits are not targets.

COMPANY LIMITS

The actual Domestic Airlines wind limits for each fleet — crosswind by runway condition, tailwind, gusts, limits for the first officer as pilot flying and for low-visibility / autoland — are in the **OM B**, the **FCOM / AOM** and the **AFM**. Always use them; the figures in this course are examples only.

KEY TAKEAWAYS

- Use the most restrictive limit: aircraft, company, crew, runway condition, LVO, aerodrome.
- Calculate with the gust; apply the FCOM / AOM speed additive.
- Above a limit: wait, change runway or divert.

KNOWLEDGE CHECK

Which crosswind limit applies when the first officer is pilot flying on a wet runway?

A. Always the AFM demonstrated value

B. The most restrictive applicable limit in the OM B

C. The tailwind limit

D. No limit

Answer: B. Apply the most restrictive company limit that applies to the situation.

Training material only. It does not replace the Aircraft Flight Manual, FCOM, QRH, MEL or the Domestic Airlines Operations Manual, which always take precedence.